

1. sampling 采样

Outline

- Sampling is a general procedure to generate **DT signals** from **CT signals**, where information of the original signals can be kept
- Core sampling theory:**
 - Impulse train, zero-order hold, 1st-order hold, etc.
 - Analysis in frequency domain
 - Nyquist rate
- Undersampling:** Aliasing
- Application:** process continuous-time signals discretely
- More sampling techniques:** decimation, downsampling and upsampling

① 如何采样 → 脉冲序列
零阶保持
一阶保持

② 如何保证能恢复出原有信号
→ 采样定理

③ 升、降采样

1.1 采样过程和原理

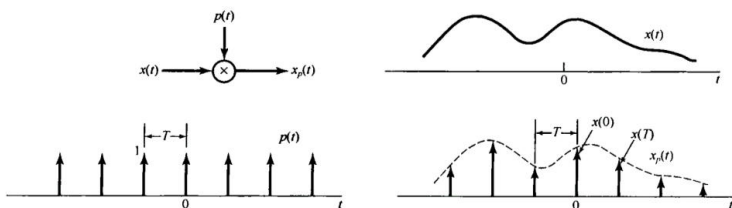


图 7.2 冲激串采样 时域

- Principle:

$$x(t) \times p(t) \Rightarrow \frac{1}{2\pi} X(j\omega) * P(j\omega)$$

- Fourier series of $p(t)$:

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \sum_{n=-\infty}^{\infty} \delta(t - nT) e^{-jk\omega_s t} dt \quad \text{where } \omega_s = \frac{2\pi}{T}$$

$$= \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_s t} dt = \frac{1}{T}$$

- Fourier Transform of $p(t)$:

$$P(j\omega) = 2\pi a_k \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

- Fourier transform of sampled signal $x_p(t)$:

$$X_p(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta) P(j(\omega - \theta)) d\theta$$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$$

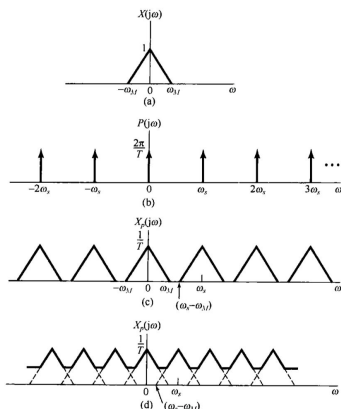


图 7.3 时域采样在频域中的效果。(a) 原始信号频谱；(b) 采样函数的频谱；(c) $\omega_s > 2\omega_m$ 时已采样信号的频谱；(d) $\omega_s < 2\omega_m$ 时已采样信号的频谱

1.2 采样定理

Sampling Theorem

Let $x(t)$ be a band-limited signal with

$$X(j\omega) = 0 \text{ for } |\omega| > \omega_M.$$

Then, $x(t)$ is uniquely determined by its samples $x(nT)$ or $x_p(t)$ if

$$\omega_s = \frac{2\pi}{T} > 2\omega_M,$$

where $2\omega_M$ is referred to as the **Nyquist rate**.

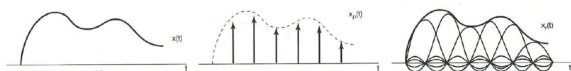
采样定理:

设 $x(t)$ 是某一个带限信号, 在 $|\omega| > \omega_M$ 时, $X(j\omega) = 0$ 。如果 $\omega_s > 2\omega_M$, 其中 $\omega_s = 2\pi/T$, 那么 $x(t)$ 就唯一地由其样本 $x(nT)$, $n = 0, \pm 1, \pm 2, \dots$ 所确定。

已知这些样本值, 我们能用如下办法重建 $x(t)$: 产生一个周期冲激串, 其冲激幅度就是这些依次而来的样本值; 然后将该冲激串通过一个增益为 T , 截止频率大于 ω_M 而小于 $\omega_s - \omega_M$ 的理想低通滤波器, 该滤波器的输出就是 $x(t)$ 。

1.3 信号重建恢复方法 插值 interpolation

Time domain interpretation of lowpass filtering



$$\begin{aligned} x_r(t) &= x_p(t) * h(t) = \sum_{n=-\infty}^{+\infty} x(nT)h(t-nT) \\ &= \sum_{n=-\infty}^{+\infty} x(nT) \frac{\sin \frac{\omega_s}{2}(t-nT)}{\frac{\omega_s}{2}(t-nT)} = \sum_{n=-\infty}^{+\infty} x(nT) \text{sinc}\left(\frac{t-nT}{T}\right) \end{aligned}$$

Ideal lowpass filtering: interpolation with sinc function

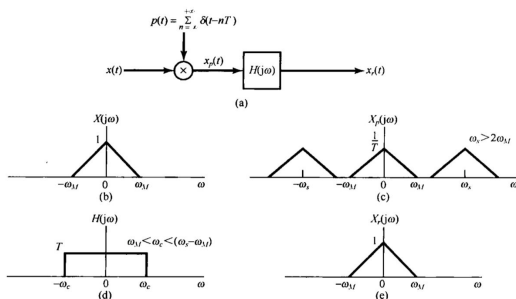


图 7.4 利用一个理想低通滤波器从信号的样本中完全恢复一个连续时间信号。(a) 采样与恢复系统; (b) $x(t)$ 的频谱; (c) $x_p(t)$ 的频谱; (d) 用于从 $X_p(j\omega)$ 恢复 $X(j\omega)$ 的理想低通滤波器; (e) $x_r(t)$ 的频谱

1.4 零阶保持 zero-order hold

- Alternative approach: zero-order hold

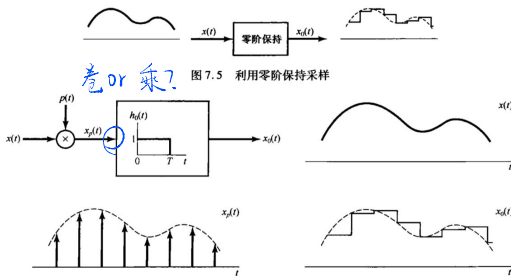
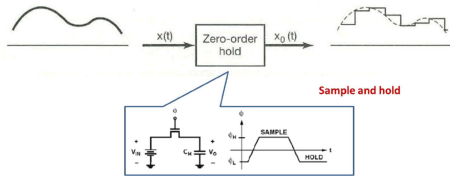


图 7.6 作为冲激串采样，再紧跟一个具有矩形单位冲激响应的线性时不变系统的零阶保持

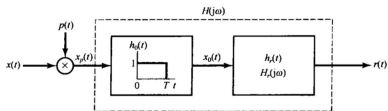


图 7.7 零阶保持 (见图 7.6) 与一个重建滤波器的级联

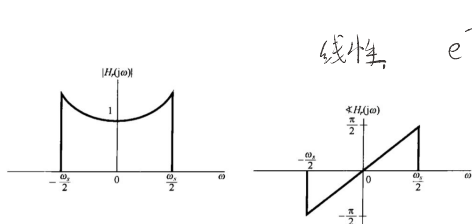
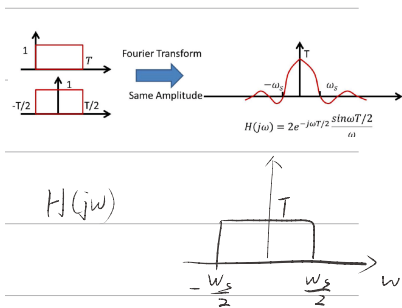
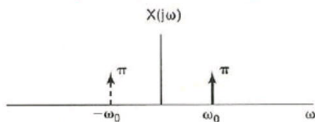
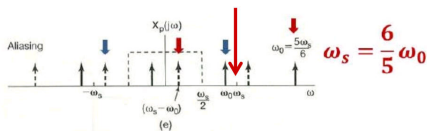
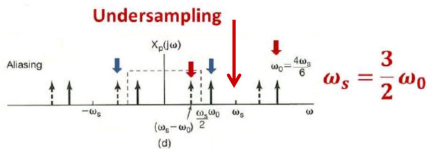


图 7.8 为零阶保持采样重建信号的重建滤波器的模和相位特性



1.5 混叠 Aliasing





Aliasing: $\cos(\omega_s - \omega_0)t$

1.6 CT - DT - CT

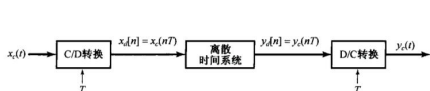
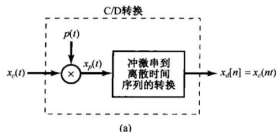


图 7.20 连续到离散时间转换和离散到连续时间转换的概念。T 代表采样周期



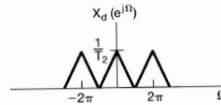
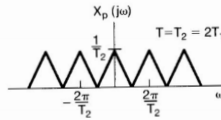
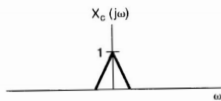
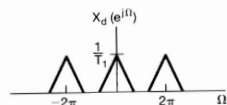
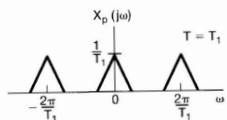
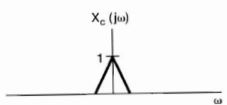
$$x_p(t) = \sum_{n=-\infty}^{+\infty} x_c(nT)\delta(t - nT)$$

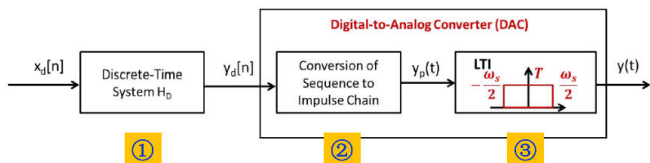
$$X_p(j\omega) = \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\omega nT}$$

$$\begin{aligned} X_d(e^{j\Omega}) &= \sum_{n=-\infty}^{+\infty} x_d[n]e^{-j\Omega n} \\ &= \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\Omega n} \end{aligned}$$

$\Rightarrow X_d(e^{j\Omega}) = X_p(j\Omega/T)$

拉伸 T 倍





• Mathematical Interpretation (Fourier Transform)

① $y_d[n] = x_d[n] * h_D[n] \longleftrightarrow Y_d(e^{j\omega}) = X_d(e^{j\omega})H_D(e^{j\omega})$

② $y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT) \longleftrightarrow Y_p(j\omega) = Y_d(e^{j\omega T})$

③ $y(t) = y_p(t) * h_{LP}(t) \longleftrightarrow Y(j\omega) = Y_p(j\omega)H_{LP}(j\omega)$

$$Y(j\omega) = X_d(e^{j\omega T})H_D(e^{j\omega T})H_{LP}(j\omega)$$

$$Y(j\omega) = X_d(e^{j\omega T})H_D(e^{j\omega T})H_{LP}(j\omega)$$

$$X_d(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - k\omega_s))$$

$$H_c(j\omega) = \frac{Y(j\omega)}{X_c(j\omega)}$$

整系统响应

$$Y(j\omega) = \left[\frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s)) \right] H_D(e^{j\omega T})H_{LP}(j\omega)$$

$$= X_c(j\omega)H_D(e^{j\omega T})$$

$$= X_c(j\omega)\tilde{H}_D(e^{j\omega T})$$

$$H_D(e^{j\omega T})$$

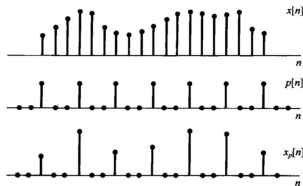
$$\Rightarrow H_c(j\omega) = \tilde{H}_D(e^{j\omega T}) = \begin{cases} H_D(e^{j\omega T}) & |\omega| < \omega_s/2 \\ 0 & \text{otherwise} \end{cases}$$

周期 $\frac{2\pi}{T}$

1.7 DT Sampling

$$x[n] \xrightarrow{\times} x_p[n]$$

$$p[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$$



$$x_p[n] = x[n]p[n] = \sum_{k=-\infty}^{+\infty} x[kN]\delta[n - kN]$$

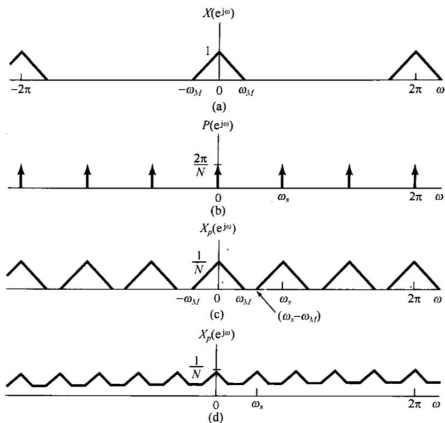
时域乘

$$X_p(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P(e^{j\theta})X(e^{j(\omega-\theta)})d\theta$$

频域卷

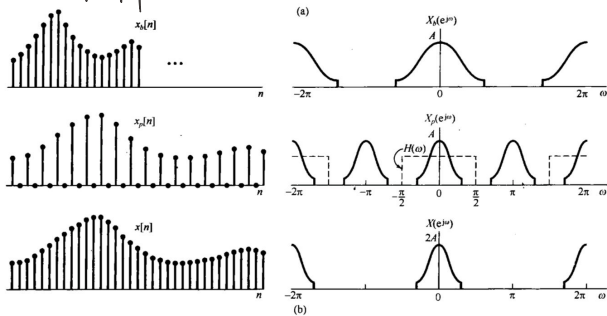
$$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

$$X_p(e^{j\omega}) = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega - k\omega_s)})$$



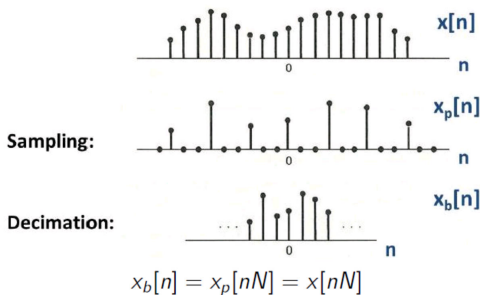
1.8 Upsampling Down sampling

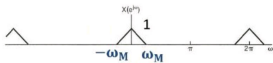
4 采样



插零 $\Rightarrow$ 压缩
+ 频带
LPF

降采样 抽取 Decimation

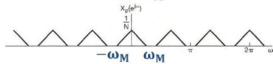




$$\omega_s = 2\pi$$

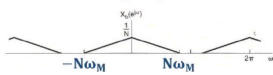
$$\omega_s = \frac{2\pi}{N}$$

Sampling:



$$\omega_s = \frac{2\pi}{N}$$

Decimation:



$$\omega_s = 2\pi$$

Condition for Perfect Reconstruction: $N\omega_M < \pi$

外生质:

Time Expansion

► Define $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k, \\ 0, & \text{Otherwise} \end{cases}$, then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega})$$